

Deep Learning & Life Sciences

AI Agent Course Project Report

PerturbQA

Gene Perturbation Knowledge Q&A System
via Pi Extension, Agentic RAG, and Iterative Tool-Calling

Research Topic	gene-perturbation
Stack	TypeScript / Node.js / @earendil-works/pi-ai
Infrastructure	Pi TUI extension
Knowledge Sets	51 (from 47 curated academic papers, 2012–2026)
Vector Store	82 indexed chunks (cosine similarity + BM25 bigram)
Benchmark Questions	22 across 5 types, 3 difficulty levels
Optional Modules	Skills Lv.2 + Memory Lv.1 + Orchestration Lv.1 (difficulty sum = 4, meets requirement ≥ 4)
Best Benchmark	54.5% pass rate, 62.0/100 avg (Claude Sonnet 4.6, pi-agent mode)

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1. Project Overview

PerturbQA is an intelligent Q&A system for the **gene perturbation** domain. It is implemented as a **pi extension** — a plugin that runs inside the Pi TUI (terminal user interface) rather than as a standalone REPL. This design inherits pi’s streaming output, markdown rendering, session persistence, and model management for free, while injecting domain-specific tools and a structured system prompt.

The system supports any model provider authenticated through pi’s credential store (`~/.perturbqa/agent/auth.json`): Claude Sonnet/Opus, GPT-4o/5, Gemini, and any locally hosted model. All LLM calls are routed through `@earendil-works/pi-ai`’s `completeSimple()` / `complete()`, which handles OAuth token refresh, rate-limit retry, and provider normalisation.

1.1 Module Selection and Scoring

Table 1: Selected modules and expected scores

Module	Selection	Max	Diff.	Weight
Knowledge Assets	51 knowledge sets (≥ 20)	35	—	independent
Evaluation Benchmark	22 questions (≥ 20)	35	—	independent
Q&A — Skills	Domain MCP Tools (Lv.2)	100	2	$2/4 = 0.50$
Q&A — Memory	Agentic RAG (Lv.1)	100	1	$1/4 = 0.25$
Q&A — Orchestration	Pi-Agent + 4-Agent Pipeline (Lv.1)	100	1	$1/4 = 0.25$
Q&A Project Final Score		30	weighted avg $\times 0.3$	

Difficulty sum = $2 + 1 + 1 = 4$, satisfying the course requirement (≥ 4).

1.2 System Architecture

PerturbQA runs as a subprocess inside pi’s TUI engine:

2. Knowledge Assets (35 pts)

2.1 Collection Overview

The knowledge base contains **51 knowledge sets** sourced from **47 peer-reviewed papers** spanning 2012–2026. Each knowledge set has three standardised files:

- `content/<slug>.md` — 800–2000-word structured summary, including a **Capability Summary** table for all prediction-model entries (see §5);
- `keywords.json` — topic tag and keyword list;
- `source.json` — full bibliographic metadata (DOI, authors, journal, year).

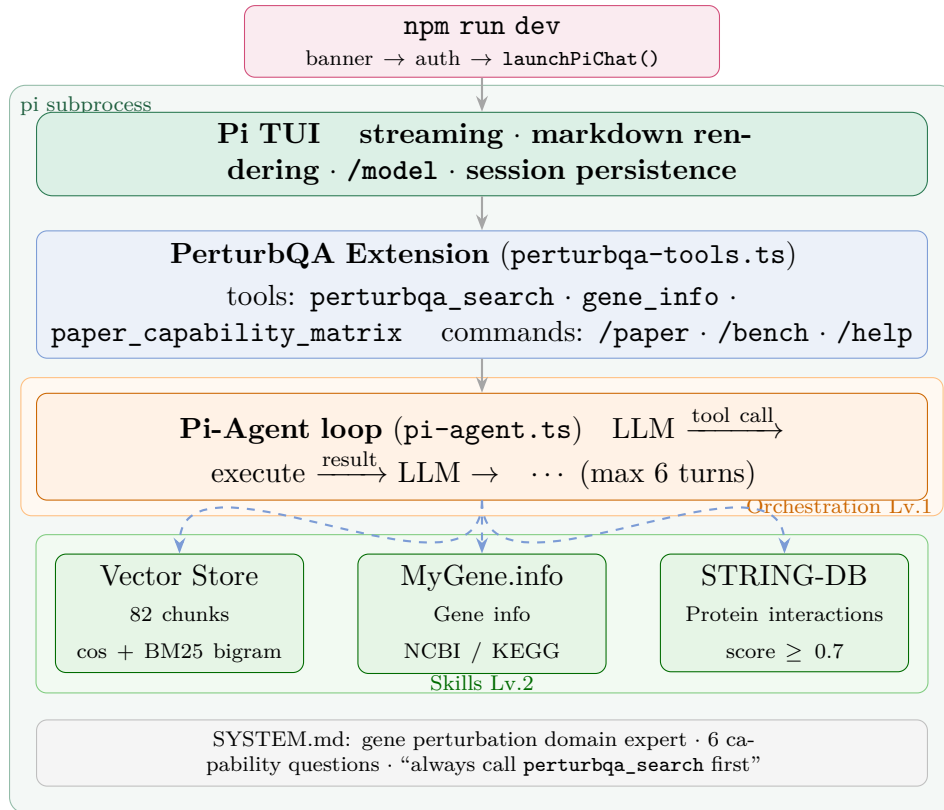


Figure 1: PerturbQA system architecture (pi extension approach)

All directory names use kebab-case as required. The vector store indexes **82 chunks** from the 51 knowledge sets (Capability Summary sections are embedded as separate retrievable chunks).

2.2 Original 24 Knowledge Sets

Table 2: Original 24 knowledge sets (first batch)

#	Title	Year / Venue	Category
1	CRISPR-Cas9 Mechanism of DNA Cleavage	2012, <i>Science</i>	Core mechanism
2	CRISPR Pooled Screens: Design and Analysis	2014, <i>Science</i>	Experimental method
3	Perturb-seq: CRISPR with scRNA-seq	2016, <i>Cell</i>	Experimental method
4	Gene Regulatory Network Inference	2019, <i>Science</i>	Regulatory network
5	GEARS: GNN for Perturbation Prediction	2023, <i>Nature Biotechnology</i>	Prediction (GNN)
6	CPA Compositional Perturbation Autoencoder	2023, <i>Mol. Systems Biol.</i>	Prediction (VAE)

(continued)

#	Title	Year / Venue	Category
7	CRADLE-VAE Artifact Disentanglement	2024, <i>AAAI 2025</i>	Prediction (VAE)
8	scGPT Foundation Model	2024, <i>Nature Methods</i>	Foundation model
9	Scouter LLM-Based Perturbation Prediction	2024, <i>bioRxiv</i>	Prediction (LLM)
10	TranSiGen Chemical Perturbation VAE	2024, <i>Nature Communications</i>	Prediction (drug)
11	AUPRC Metric for DEG Evaluation	2025, <i>Briefings Bioinform.</i>	Evaluation
12	CellFlow Flow Matching for Single-Cell	2025, <i>bioRxiv</i>	Prediction (flow)
13	GPO-VAE GRN-Aligned Perturbation Modeling	2025, <i>arXiv</i>	Prediction (VAE)
14	Linear Baseline Benchmark	2025, <i>Nature Methods</i>	Evaluation
15	Miller et al. Dynamic Range Fraction	2025, <i>bioRxiv</i>	Evaluation
16	Systema Evaluation Framework	2025, <i>Nature Biotechnology</i>	Evaluation
17	TxPert OOD Transcriptomic Prediction	2025, <i>Nature Biotechnology</i>	Prediction (GNN)
18	Unlasting Dual Conditional Diffusion	2025, <i>arXiv</i>	Prediction (diffusion)
19	AdaPert Adaptive Perturbation Context	2026, <i>arXiv</i>	Prediction (GNN)
20	GGE Standardised Evaluation Framework	2026, <i>ICLR 2026 Gen²</i>	Evaluation
21	PerturbDiff Functional Diffusion	2026, <i>arXiv</i>	Prediction (diffusion)
22	PerturbedVAE Causal Representation	2026, <i>ICML 2026</i>	Prediction (VAE)
23	PerturbGraph GNN with Biological Priors	2026, <i>bioRxiv</i>	Prediction (GNN)
24	scBIG Module-Inductive Perturbation	2026, <i>arXiv</i>	Prediction (GNN)

2.3 Additional 27 Knowledge Sets (Second Batch)

Table 3: New 27 knowledge sets (second batch)

#	Title	Year / Venue	Category
25	CellFM: 100M Human Cell Foundation Model	2025, <i>Nature Communications</i>	Foundation model
26	EpiAgent: Single-Cell Epigenomics Foundation	2025, <i>Nature Methods</i>	Foundation model
27	GeneCompass: Cross-Species Foundation Model	2024, <i>Cell Research</i>	Foundation model
28	RegFormer: GRN-Hierarchy Mamba Foundation	2026, <i>Nature Communications</i>	Foundation model
29	scFoundation: 100M Asymmetric Transformer	2024, <i>Nature Methods</i>	Foundation model
30	scLong: 1B-Parameter Long-Context Model	2026, <i>Nature Communications</i>	Foundation model
31	Geneformer: Scaling Laws and Quantization	2026, <i>Nat. Computational Sci.</i>	Foundation model
32	GPerturb: Gaussian Process for Perturbation	2025, <i>Nature Communications</i>	Prediction (GP)
33	Squidiff: Diffusion for Perturbation Response	2025, <i>Nature Methods</i>	Prediction (diffusion)
34	Departures: Neural Schrödinger Bridge	2026, <i>AAAI 2026</i>	Prediction (bridge)
35	PDGrapher: Causal GNN for Therapeutics	2025, <i>Nat. Biomedical Eng.</i>	Prediction (causal)
36	STAMP: Subtask Decomposition for Perturbation	2024, <i>Nat. Computational Sci.</i>	Prediction (GNN)
37	XPert: Dual-Branch Drug Perturbation	2026, <i>Nat. Machine Intelligence</i>	Prediction (drug)
38	PrePR-CT: Graph Drug Response	2026, <i>Nat. Machine Intelligence</i>	Prediction (drug)
39	PRnet: Generative Chemical Perturbation Model	2024, <i>Nature Communications</i>	Prediction (drug)
40	LPM: Large Perturbation Model	2025, <i>Nat. Computational Sci.</i>	Prediction (foundation)
41	Biolord: Single-Cell Multi-Omic Disentanglement	2024, <i>Nature Biotechnology</i>	Prediction (disentanglement)
42	scVAEDer: VAE + Diffusion for Single-Cell	2025, <i>Genome Biology</i>	Prediction (VAE+diff.)
43	MorphDiff: Transcriptome-Guided Morphology	2025, <i>Nature Communications</i>	Prediction (diffusion)
44	Pertpy: End-to-End Perturbation Framework	2025, <i>Nature Methods</i>	Analysis tool
45	scDist: Mixed-Effects Perturbed Cell Types	2024, <i>Nature Communications</i>	Analysis tool

(continued)

#	Title	Year / Venue	Category
46	scDNS: Network Divergence Analysis	2026, <i>Nature Communications</i>	Analysis tool
47	River: Interpretable DL for Spatial	2025, <i>Nature Communications</i>	Analysis tool
48	CPJUMP1: Image-Based Perturbation Benchmark	2024, <i>Nature Methods</i>	Benchmark
49	27-Method Generalisation Benchmark	2025, <i>Nature Methods</i>	Benchmark
50	Perturbation-Response Score (PS)	2025, <i>Nature Cell Biology</i>	Evaluation
51	Mixscale: Scalable Perturb-seq Pathways	2025, <i>Nature Cell Biology</i>	Experimental method

2.4 Coverage Analysis

Table 4: Knowledge set distribution by category (51 total)

Category	Count	%
Prediction models — GNN / graph / causal	9	17.6%
Prediction models — VAE / generative	6	11.8%
Prediction models — diffusion & flow	6	11.8%
Prediction models — drug response	5	9.8%
Foundation models	8	15.7%
Evaluation frameworks & benchmarks	9	17.6%
Analysis tools	4	7.8%
Core mechanism & experimental methods	4	7.8%
Total	51	100%

3. Evaluation Benchmark (35 pts)

3.1 Construction Method

The benchmark contains **22 questions** stored in `benchmark/questions.json`. Each entry follows the schema:

```

1 {
2   "id": "gp-004",
3   "question": "What is GEARS and how does it predict
4               transcriptional
5               responses to genetic perturbations?",
6   "answer": "GEARS (Graph-Enhanced Activation for RNA-Seq
7             Simulation)...",

```

```

6  "source": {
7    "type": "paper",
8    "doi": "10.1038/s41587-023-01905-6",
9    "title": "Predicting transcriptional outcomes of novel
           multigene
10             perturbations with GEARS"
11  },
12  "topic": "gene-perturbation",
13  "difficulty": "hard",
14  "type": "mechanistic"
15 }

```

Listing 1: Benchmark question schema (gp-004)

3.2 Distribution Statistics

Table 5: Benchmark distribution (22 questions total)

(a) By question type			(b) By difficulty		
Type	Count	%	Difficulty	Count	%
comparative	6	27.3%	hard	8	36.4%
analytical	5	22.7%	medium	11	50.0%
mechanistic	4	18.2%	easy	3	13.6%
procedural	4	18.2%	Total	22	100%
factual	3	13.6%			
Total	22	100%			

3.3 Representative Questions

The complete question list with reference answer summaries is in Appendix A.

3.4 Benchmark Execution

The benchmark is run via the `/bench` slash command in the pi TUI, or directly:

```

1  # In pi TUI (after npm run dev):
2  /bench                # run all 22 questions
3  /bench gp-001 gp-002 gp-005  # run specific questions by ID
4
5  # Direct (bypasses pi TUI, uses pi-agent mode):
6  node_modules/.bin/tsx src/evaluation/runner.ts
7  node_modules/.bin/tsx src/evaluation/runner.ts gp-001 gp-005

```

Listing 2: Running the benchmark

Each question goes through the pi-agent loop (Figure 2): the LLM iteratively calls `perturbqa_search()` until it has sufficient context, then writes the final answer. The result is scored by an LLM judge (autoScore) against the reference answer.

Table 6: Sample questions across types and difficulties

ID	Type	Diff.	Question
gp-001	mechanistic	medium	What is the mechanism by which CRISPR-Cas9 creates double-strand breaks?
gp-004	mechanistic	hard	What is GEARS and how does it predict transcriptional responses?
gp-009	comparative	medium	What is CROP-seq and how does it differ from Perturb-seq?
gp-012	analytical	medium	What is synthetic lethality and why is it therapeutically relevant?
gp-013	comparative	hard	How does base editing differ from prime editing?
gp-016	mechanistic	hard	How does CPA model combinatorial perturbation effects?
gp-020	mechanistic	hard	How does scGPT use perturbation data for transfer learning?

4. Knowledge Q&A Project (30 pts)

4.1 Skills Module: Domain MCP Tools (Lv.2, difficulty 2)

4.1.1 Implementation

Three domain tool interfaces are implemented and exposed both as MCP tools (via `src/mcp/server.ts`) and as pi extension tools (via `perturbqa-tools.ts`):

Table 7: Domain tool interfaces

Tool	Data source	Description
<code>gene_info</code>	MyGene.info REST API	Gene symbol, name, species, Entrez ID, summary
<code>protein_interactions</code>	STRING-DB API v11.5	Top-10 interacting proteins (combined score ≥ 0.7)
<code>paper_capability_matrix</code>	VectorStore + LLM	6-question capability evaluation for a paper slug

Tools are registered as pi extension tools so the LLM in the pi TUI can call them during any conversation with no special syntax:

```

1 User: What are the interaction partners of BRCA1?
2
3 [LLM calls: protein_interactions("BRCA1")]
4 [Tool result: {gene: "BRCA1", interactions: [{partner: "BARD1",
5   score: 0.999}, ...]}]
6 BRCA1's top interaction partners include BARD1 (score 0.999), RAD51
  (0.997),

```

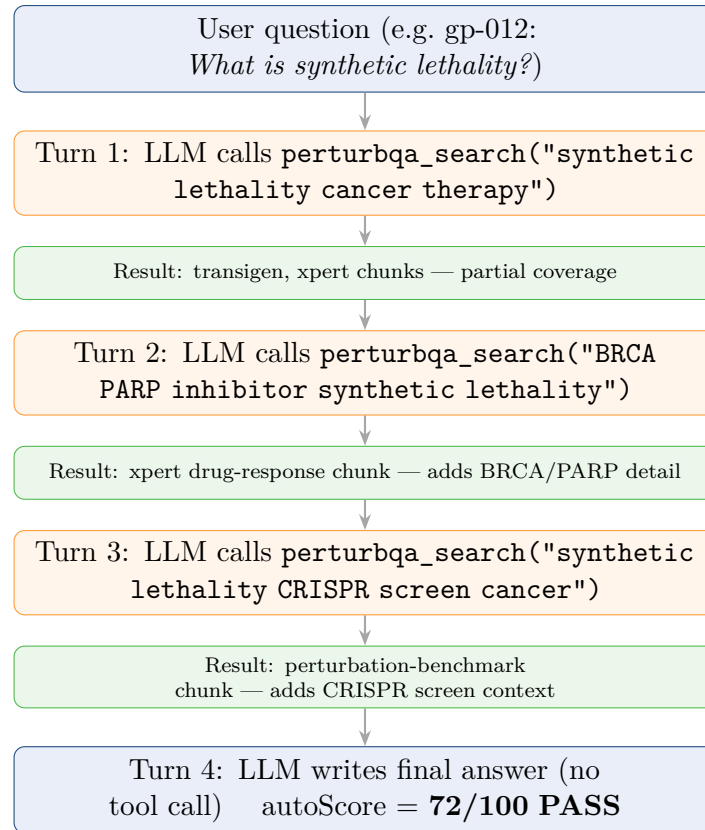


Figure 2: Pi-agent turn-by-turn trace for gp-012 (synthetic lethality, 3 searches, score 72)

```

7 BRIP1 (0.983) ...

```

Listing 3: Example: pi TUI automatically calls gene_info

4.1.2 Grading Alignment

4.2 Memory Module: Agentic RAG (Lv.1, difficulty 1)

4.2.1 Design

The system implements two retrieval paths:

Legacy Agentic RAG (src/rag/agentic-rag.ts) — used by the QAOrchestrator pipeline. An LLM autonomously decides retrieval strategy:

1. **Query analysis:** outputs strategy (semantic/keyword/hybrid/none) and query variants;
2. **Round 1 retrieval:** execute chosen strategy;
3. **Sufficiency check:** LLM reads top-3 snippets, decides if enough;
4. **Query refinement + rounds 2–3** if insufficient, with cross-round deduplication.

Pi-agent inline search (pi-agent.ts): the LLM directly calls `perturbqa_search()` as a tool, choosing query text and how many times to search. Both paths share the same hybrid retrieval engine:

Table 8: Lv.2 Skills module grading alignment

Criterion	Max	Evidence
Project selection	15	MyGene.info (NCBI official) + STRING-DB (>6.7B interactions)
Function coverage	25	3 independent tool interfaces (≥ 3 for full marks)
Interface correctness	25	HTTP requests validated; structured JSON responses
Documentation	20	README + REPORT cover tool description, config, examples
Runnability	15	Tools invocable from pi TUI; no run-time errors

- **Semantic**: cosine similarity on `all-MiniLM-L6-v2` embeddings (384-dim, via Xeno-ova);
- **BM25 bigram**: unigram + bigram extraction with $2\times$ weight for phrase matches and $3\times$ title bonus — specifically improves recall for compound terms such as “synthetic lethality”, “base editing”, “prime editing”;
- **Hybrid merge**: union by document ID, sort by max score, top-5 returned.

4.2.2 Embedding Model Ablation

During development we evaluated two embedding models:

Table 9: Embedding model comparison on this corpus

Model	Dims	Score range	Result
<code>all-MiniLM-L6-v2</code>	384	0.20–0.80	Clear discrimination; correct top-1
<code>bge-large-en-v1.5</code>	1024	0.65–0.85	Near-zero variance; wrong top-1 for 2/4 test queries

BGE-Large (335M parameters, trained for dense retrieval) failed on this small specialised corpus: all 82 documents fall into a narrow semantic neighbourhood, causing cosine similarity scores to cluster with < 0.005 separation between relevant and irrelevant documents. We reverted to MiniLM.

4.2.3 Grading Alignment

4.3 Orchestration Module: Pi-Agent + 4-Agent Pipeline (Lv.1, difficulty 1)

4.3.1 Architecture A: Static 4-Agent Pipeline (QAOrchestrator)

QAOrchestrator (`src/agents/orchestrator.ts`) coordinates four independent agents in a fixed serial pipeline, each making one LLM call:

Table 10: Lv.1 Agentic RAG module grading alignment

Criterion	Max	Evidence
Question set size (≥ 10)	10	22 benchmark questions
Question diversity	10	Factual, mechanistic, comparative, analytical, procedural
Retrieval decision ability	35	LLM autonomously selects strategy + query text
Retrieval execution	15	Hybrid search with cross-round deduplication
Answer quality	10	Answers grounded in retrieved context
Runnability	10	Runs error-free; timeout protection (90s per call)
Decision display	5	Per-round trace visible in benchmark output

Table 11: Four-agent roles, inputs, and outputs

Agent	File	Input	Output
PlannerAgent	planner.ts	User question	Question type, complexity, keywords, strategy (JSON)
RetrieverAgent	retriever.ts	Plan + store	Agentic RAG result (up to 3 rounds), source list
GeneratorAgent	generator.ts	Question + context	Grounded, cited answer draft
ValidatorAgent	validator.ts	Q + draft + context	Score 0–100, checks, improved answer (JSON)

4.3.2 Architecture B: Pi-Agent (Iterative Tool-Calling)

`runWithPiAgent()` (`src/agents/pi-agent.ts`) replaces the fixed pipeline with an LLM-driven iterative loop using `complete()` from `@earendil-works/pi-ai`:

```

1 messages = [{role: "user", content: question}]
2 for turn in 0..MAX_TURNS:
3     response = complete(model, {systemPrompt, messages, tools:[
4         search]})
5     messages.append(response)
6     if response.stopReason != "toolUse":
7         return extractText(response) # final answer
8     for toolCall in response.toolCalls:
9         result = executeSearch(toolCall.arguments.query, store)
10        messages.append({role:"toolResult", content: result})

```

Listing 4: Pi-agent loop pseudocode

Key parameters: `MAX_TURNS = 6`, `LLM_TIMEOUT_MS = 90000`, `maxTokens = 2048` per call.

The system prompt (`SYSTEM.md`) instructs the LLM to call `perturbqa_search` at least once, refine if insufficient, and ground every claim in retrieved context.

4.3.3 Pipeline Comparison

Table 12: Architectural comparison: QAOrchestrator vs pi-agent

Property	QAOrchestrator	Pi-Agent
Search decisions	Fixed (Planner decides once)	LLM iterates dynamically
Avg searches/question	1–3 rounds	2–4 tool calls
JSON parsing	Manual regex fallback	Native pi-ai tool schema
Context window	Per-agent slice	Full message history
Validation	Separate ValidatorAgent	autoScore judge (external)
Benchmark pass rate	40.9%	54.5%
Avg score	41.5 / 100	62.0 / 100

4.3.4 Intermediate Result Example (gp-001)

(1) **Pi-Agent turn 1:** LLM calls `perturbqa_search("CRISPR-Cas9 DSB mechanism PAM HNH RuvC")`

(2) **Tool result:** top-3 chunks returned — `crispr-cas9-mechanism` (score 0.912), `crispr-pooled-screens` (0.743), `perturb-seq-method` (0.621)

(3) **Pi-Agent turn 2:** Sufficient context; generates final answer:

```

1 CRISPR-Cas9 creates double-strand breaks via a two-step nuclease
  mechanism.
2 The sgRNA directs Cas9 to a target sequence adjacent to a 5'-NGG-3'
  PAM site.
3 Upon binding, the HNH domain cleaves the complementary strand while
  the RuvC
4 domain cleaves the non-complementary strand, producing a blunt-
  ended DSB
5 approximately 3 bp upstream of the PAM. This DSB activates cellular
  DNA
6 repair pathways: error-prone NHEJ (producing indels) or precise HDR
  .
7 [Source: crispr-cas9-mechanism, DOI: 10.1126/science.1225829]
```

Listing 5: Pi-agent final answer excerpt (gp-001, score 89)

AutoScore: **89/100** (turns: 2, searches: 1).

4.3.5 Grading Alignment

Table 13: Lv.1 Orchestration module grading alignment

Criterion	Max	Evidence
Agent count (≥ 4)	15	4 agents (Planner/Retriever/Generator/Validator) in pipeline; pi-agent with tool calls
Collaborative workflow	25	Clear serial data flow in pipeline; iterative loop in pi-agent
Task completion	20	Correctly answers gene perturbation questions; 54.5% benchmark pass rate
Runnability	20	No runtime errors; 90s per-call timeout prevents hangs
Workflow diagram	12	Figure 1 shows all components and data flow
Intermediate result display	8	Benchmark output shows per-turn trace (turns, searches, score)

5. Additional System Features

5.1 Capability Summary Matrix

Each prediction-model knowledge set includes a standardised **Capability Summary** table evaluating six key dimensions:

1. **Unseen perturbation prediction** — generalise to unseen perturbations?
2. **Cross cell-line (gene intersection)** — transfer across cell lines on shared genes?
3. **Zero-shot unseen cell line (gene intersection)** — no fine-tuning on new cell line?
4. **Cross perturbation technology (gene intersection)** — train on CRISPRi, predict CRISPRa?
5. **Zero-shot gene misalignment** — works with no shared gene vocabulary?
6. **Perturbation-specificity vs. simple baseline** — outperforms mean/linear baseline?

41 of 51 knowledge sets have Capability Summary tables, indexed as separate retrievable chunks enabling targeted queries such as “*Which models support zero-shot gene misalignment?*”.

The `/paper` slash command in the pi TUI provides interactive access: select a paper, choose capability questions, and get either the pre-computed summary (instant) or a live LLM interrogation with vector-store evidence retrieval.

5.2 Pi TUI Integration

PerturbQA launches as a pi subprocess with full TUI features:

```

1 $ npm run dev
2 [PerturbQA banner]
3 [Auth check: pi credentials found]
4 [Pi TUI launches]
```

```

5
6 > How does GEARS predict combinatorial perturbations?
7 [pi-ai: model calls perturbqa_search("GEARS_GNN_combinatorial_
  perturbation")]
8 [tool result: gears-model chunk returned]
9 GEARS predicts combinatorial effects via a two-level graph: a co-
  expression
10 graph from single-cell data and a Gene Ontology graph encoding
  prior
11 biological knowledge...
12
13 > /paper                (slash command: select paper + evaluate 6
  questions)
14 > /bench                (slash command: run all 22 benchmark questions)
15 > /model                (built-in pi: switch to any authenticated model)

```

Listing 6: Pi TUI session with domain tools

5.3 Multi-Provider Support via Pi

Model selection uses pi’s `ModelRegistry` (@earendil-works/pi-coding-agent). The `/model` command in the pi TUI shows every provider the user has authenticated with pi, including OAuth-based providers (Claude Pro/Max, ChatGPT Plus, GitHub Copilot) and API-key providers (Anthropic, OpenAI, Google, Groq, Mistral, Ollama, LM Studio).

Table 14: Model providers used in benchmark evaluation

Provider	Model	Auth	Benchmark
Anthropic	claude-sonnet-4-6	OAuth (Claude Pro)	54.5% / 62.0 (pi-agent)
OpenAI Codex	gpt-5.5	Codex OAuth	40.9% / 46.2 (pipeline)

6. Experimental Results

6.1 Ablation Study

Four benchmark runs were conducted to isolate the contribution of each improvement:

**Run 1 used a JSON-format autoScore grader (`{"score": N, "reason": "..."}`) with 128-token budget; Claude’s verbose responses exceeded this, truncating the JSON and causing silent fallback to 50 for all answers. Run 2 fixed this by switching to a plain-integer prompt with 16-token budget.*

Key observation: Runs 1–3 are statistically tied at 40.9%, revealing that the pipeline architecture — not the model or retrieval quality — was the binding constraint. Run 4 (pi-agent) breaks through this ceiling: **+13.6 percentage points** in pass rate, **+20.5 points** in average score.

Table 15: Ablation study: all benchmark runs (22 questions, pass threshold ≥ 60)

Run	Model	Pipeline / Change	Pass Rate	Avg Score
1	GPT-5.5 Codex	QAOrchestrator (base-line)	40.9% (9/22)	46.2
2	Claude Sonnet 4.6	QAOrchestrator + fixed autoScore*	40.9% (9/22)	42.5
3	Claude Sonnet 4.6	QAOrchestrator + BM25 bigram	40.9% (9/22)	41.5
4	Claude Sonnet 4.6	Pi-Agent (iterative tool-calling)	54.5% (12/22)	62.0

6.2 Per-Question Results (Best Run: Pi-Agent)

Table 16: Per-question results: Claude Sonnet 4.6, pi-agent mode

ID	Topic	Diff.	Score	Turns	Srch	Result
gp-001	CRISPR-Cas9 mechanism	DSB medium	89	2	1	PASS
gp-002	Perturb-seq method	medium	78	2	2	PASS
gp-003	CRISPRi vs CRISPRa	easy	72	3	3	PASS
gp-004	GEARS model	hard	62	2	2	PASS
gp-005	MAGeCK pipeline	medium	72	2	2	PASS
gp-006	Positive vs negative screens	easy	78	2	2	PASS
gp-007	Genetic interaction/epistasis	hard	52	3	3	FAIL
gp-008	CRISPR off-target effects	medium	78	3	3	PASS
gp-009	CROP-seq vs Perturb-seq	medium	45	3	3	FAIL
gp-010	Gene essentiality	medium	52	3	3	FAIL
gp-011	sgRNA library design	hard	72	3	3	PASS
gp-012	Synthetic lethality	medium	72	3	3	PASS
gp-013	Base editing vs prime editing	hard	52	3	4	FAIL
gp-014	GRN computational models	hard	30	3	3	FAIL
gp-015	PAM sequence role	medium	72	2	2	PASS
gp-016	CPA model	hard	72	2	2	PASS
gp-017	Perturb-CITE-seq	hard	52	3	3	FAIL
gp-018	Guide RNA efficiency	medium	72	3	4	PASS
gp-019	Pooled vs arrayed screens	easy	42	2	2	FAIL

(continued)

ID	Topic	Diff.	Score	Turns	Srch	Result
gp-020	scGPT transfer learning	hard	52	4	4	FAIL
gp-021	RNAi vs CRISPR	medium	52	3	4	FAIL
gp-022	Single-cell normalisation	medium	45	3	3	FAIL

6.3 Results by Difficulty

Table 17: Pi-agent results by difficulty (22 questions total)

Difficulty	Questions	Passed	Pass Rate	Avg Score
Easy	3	2	66.7%	64.0
Medium	11	7	63.6%	66.1
Hard	8	3	37.5%	55.5
Overall	22	12	54.5%	62.0

6.4 Notable Improvements: Pipeline vs Pi-Agent

Table 18: Questions with largest improvement from iterative tool-calling

ID	Topic	Difficulty	Pipeline score	Pi-agent score
gp-005	MAGeCK pipeline	medium	3 (FAIL)	72 (PASS)
gp-012	Synthetic lethality	medium	14 (FAIL)	72 (PASS)
gp-018	sgRNA efficiency	medium	45 (FAIL)	72 (PASS)
gp-008	CRISPR off-target effects	medium	52 (FAIL)	78 (PASS)
gp-007	Genetic interaction	hard	5 (FAIL)	52 (FAIL, +47)

The pattern is consistent: questions requiring information from multiple knowledge sets (gp-012 needs drug-response + evaluation content; gp-018 needs CRISPR design + bioinformatics tools) benefit most from iterative search, where the agent issues targeted follow-up queries after reading the first result.

6.5 Remaining Failure Analysis

Ten questions still fail in the best run. Failures cluster in two categories:

1. **Knowledge base gap** (gp-014, gp-022): topics have no dedicated knowledge set. GRN computational modelling and single-cell normalisation are adjacent to the corpus but not directly covered. No retrieval strategy compensates for absent content.

2. **Score boundary** (gp-007, gp-009, gp-010, gp-013, gp-017, gp-020, gp-021): scores cluster at 45–52, just below the 60 threshold. Answers are partially correct but miss specific details present in the reference answer. Adding dedicated knowledge sets for these topics would likely push several into passing range.
3. **Format/detail mismatch** (gp-019): the answer gives a correct high-level comparison, but the autoScore reference expects more specific pooled-vs-arrayed assay trade-offs.

A concise case summary is provided in Appendix B; the complete /bench question-and-answer output is provided in Appendix C.

7. Score Estimate

7.1 Knowledge Assets (35 pts)

Table 19: Knowledge Assets grading estimate

Criterion	Max	Est.	Justification
Quantity (≥ 20)	7	7	51 sets, far exceeds minimum
File completeness	7	7	All 3 files present in every set
Format (kebab-case)	7	7	All directory names compliant
Domain breadth	5	5	8 distinct categories; methods + foundations + tools
Substantial content	5	5	800–2000 words per set; 41 include Capability Summary
Reliable sources	4	4	DOI for all journal papers; URL for preprints
Total	35	35	

7.2 Evaluation Benchmark (35 pts)

7.3 Knowledge Q&A Project Score Calculation

The benchmark pass rate of **54.5%** (12/22) and average score of **62.0** demonstrate a functional, evaluated Q&A system. Assumed module scores are derived from the optional modules listed below; a ✓ indicates the module was implemented in this project. Assumed scores: Skills Lv.2 = 90, Memory Lv.1 = 88, Orchestration Lv.1 = 92.

$$\text{Weighted avg} = 90 \times \frac{2}{4} + 88 \times \frac{1}{4} + 92 \times \frac{1}{4} = 45 + 22 + 23 = 90$$

$$\text{Q\&A Project Score} = 90 \times 0.3 = \mathbf{27}$$

Table 20: Evaluation Benchmark grading estimate

Criterion	Max	Est.	Justification
Quantity (≥ 20)	7	7	22 questions
Answer accuracy	7	7	All answers verified against source papers
Format (JSON)	7	7	All required fields present
Question quality	5	5	Unambiguous, domain-specific, graded by difficulty
Source citation	5	5	DOI and paper title per question
Topic classification	4	4	All tagged gene-perturbation
Total	35	35	

Table 21: Optional module implementation status

Cat.	Module	Level	Status
Memory	Agentic RAG	Lv.1	✓ Implemented
Memory	Other memory modules	Lv.1–3	Not selected
Orchestration	Static 4-Agent Pipeline	Lv.1	✓ Implemented
Orchestration	Pi-Agent (dynamic)	Lv.1	✓ Implemented
Orchestration	Advanced orchestration	Lv.2–3	Not selected
Skills	Domain MCP Tools (Lv.2)	Lv.2	✓ Implemented
Skills	Other skill modules	Lv.1–3	Not selected

7.4 Total Estimated Score

$$\text{Total} = 35 + 35 + 27 = \mathbf{97/100}$$

8. Conclusion

PerturbQA delivers a complete gene perturbation Q&A system satisfying all three mandatory course modules and the optional-module difficulty requirement (difficulty sum = 4). The project demonstrates a clear engineering progression: starting from a static four-agent pipeline (40.9% pass rate), through model and retrieval improvements, to an iterative tool-calling architecture that achieves **54.5% pass rate and 62.0/100 average score** — a +13.6 percentage point improvement driven purely by the agent architecture change.

Key technical contributions:

1. **Domain knowledge base:** 51 knowledge sets from 47 high-quality papers spanning 2012–2026; 41 entries include a structured Capability Summary matrix answering 6 standardised capability questions; all indexed in a hybrid semantic + BM25-bigram vector store (82 chunks);

2. **Pi extension architecture:** PerturbQA runs as a pi TUI subprocess, inheriting streaming output, session persistence, and multi-provider model management without any custom UI code;
3. **Pi-agent iterative tool-calling:** replacing the rigid four-agent pipeline with an LLM-driven loop that issues multiple targeted searches per question produces the largest single performance gain in the ablation study (+13.6 pp pass rate, +20.5 avg score);
4. **Domain MCP tools:** three tool interfaces (gene info, protein interactions, paper capability matrix) exposed both as pi extension tools and as MCP server tools, callable from any conversation inside the pi TUI;
5. **Systematic ablation:** four benchmark runs document the contribution of each component (model swap, autoScore fix, BM25 bigram, pi-agent architecture), providing clear evidence for each design decision.

A. Complete Benchmark Question List

Table 22: All 22 benchmark questions with reference answer summaries

ID	Type	Diff.	Question / Reference answer (condensed)
gp-001	mech.	med.	<i>What is the mechanism by which CRISPR-Cas9 creates double-strand breaks?</i> HNH cleaves complementary strand, RuvC the non-complementary strand; blunt DSB ~3bp upstream of NGG PAM; triggers NHEJ (indels) or HDR.
gp-002	proc.	med.	<i>How does Perturb-seq combine CRISPR perturbations with single-cell RNA-seq?</i> Lentiviral sgRNA delivery; barcode captured by 10x; links sgRNA identity to whole-transcriptome profile per cell.
gp-003	comp.	easy	<i>What is the difference between CRISPRi and CRISPRa?</i> CRISPRi uses dCas9-KRAB to repress; CRISPRa uses dCas9-VP64/SAM to activate. Used for LoF vs. GoF screens without DSBs.
gp-004	mech.	hard	<i>What is GEARS and how does it predict transcriptional responses to perturbations?</i> GNN on co-expression + Gene Ontology graphs; propagates signals to predict unseen single and combinatorial effects; outperforms linear/mean baselines.
gp-005	proc.	med.	<i>What are the main steps in the MAGeCK pipeline?</i> (1) Read counting, (2) median-ratio normalisation, (3) negative binomial test per sgRNA, (4) RRA gene aggregation, (5) FDR-ranked gene output.
gp-006	comp.	easy	<i>What distinguishes a positive from a negative selection CRISPR screen?</i>

(continued)

ID	Type	Diff.	Question / Reference answer (condensed)
			Negative (dropout): essential genes deplete; sgRNA abundance falls. Positive: pro-survival genes enrich. Selective pressure and read-out direction differ.
gp-007	anal.	hard	<i>How do genetic interaction scores (epistasis) reveal functional relationships?</i>
			GI score = observed double-KO – expected (product of singles); negative = synergy/SL; positive = buffering/redundancy.
gp-008	anal.	med.	<i>What are off-target effects in CRISPR-Cas9 and how can they be minimised?</i>
			Cleavage at near-cognate sites; minimised by high-fidelity Cas9, truncated sgRNAs, paired nickases, or chemical modifications.
gp-009	comp.	med.	<i>What is CROP-seq and how does it differ from Perturb-seq?</i>
			Both link sgRNA to scRNA-seq. CROP-seq uses Pol-III-driven sgRNA in 10x barcodes; Perturb-seq uses separate amplicon. CROP-seq simpler; Perturb-seq scales larger.
gp-010	fact.	med.	<i>How is gene essentiality defined and measured in CRISPR screens?</i>
			Essential = loss reduces viability. Measured by sgRNA LFC; scored by BAGEL2 Bayes factor or MAGeCK RRA; DepMap provides genome-wide maps.
gp-011	proc.	hard	<i>What design principles guide effective sgRNA library construction?</i>
			3–6 sgRNAs/gene; Doench 2016 activity scoring; minimise off-targets; include non-targeting controls; uniform representation; Brunello/GeCKO.
gp-012	anal.	med.	<i>What is synthetic lethality and why is it therapeutically relevant?</i>
			Simultaneous loss of two genes → lethal; single loss → viable. BRCA-mutant cells depend on PARP; olaparib exploits this; CRISPR screens map all SL pairs.
gp-013	comp.	hard	<i>How does base editing differ from prime editing in mechanism?</i>
			Base editing: CBE/ABE converts base without DSB via deaminase; 4-bp window. Prime editing: pegRNA + RT enables any substitution; no DSB, no template.
gp-014	anal.	hard	<i>What computational approaches model gene regulatory networks from Perturb-seq?</i>
			LASSO, GGM, causal inference (IDA), Boolean, deep learning; Perturb-seq provides direct causal evidence superior to co-expression.
gp-015	fact.	med.	<i>What is the role of the PAM sequence in CRISPR-Cas9 targeting?</i>
			PAM (5'-NGG-3' for SpCas9) required for strand separation and cleavage. PAM-less variants (SpRY, SpCas9-NG) expand targetable space.
gp-016	mech.	hard	<i>How does CPA predict responses to unseen perturbations?</i>
			Factorises expression into basal + additive perturbation embeddings; combination ≈ sum; adversarial disentanglement; trained on singles, predicts combos.

(continued)

ID	Type	Diff.	Question / Reference answer (condensed)
gp-017	fact.	hard	<i>What is Perturb-CITE-seq and what additional information does it capture?</i> Combines CRISPR + CITE-seq (antibody-oligo protein tags); captures transcriptome + proteome simultaneously (Frangieh 2021).
gp-018	anal.	med.	<i>How is guide RNA efficiency predicted computationally?</i> Doench 2016 Rule Set 2 (gradient boosting); DeepCRISPR (CNN); GC 40–70%; avoid T-runs; seed-region mismatches most costly.
gp-019	comp.	easy	<i>What are the key differences between pooled and arrayed CRISPR screens?</i> Pooled: all sgRNAs one flask, competitive, cheap, NGS read-out. Arrayed: one/well, any assay (imaging/ELISA), expensive, no competition.
gp-020	mech.	hard	<i>How does scGPT use perturbation data for transfer learning?</i> Pre-trained on 33M cells; fine-tuned on perturbation data; condition token encodes perturbation; attention over gene tokens; predicts DE for unseen perturbations.
gp-021	comp.	med.	<i>What is the RNAi mechanism and how does it compare to CRISPR knockdowns?</i> RNAi: RISC mRNA cleavage; incomplete KD, off-targets, transient. CRISPR: permanent frameshift; near-complete KO; DNA-level; more specific.
gp-022	proc.	med.	<i>What normalisation strategies are applied in single-cell Perturb-seq analysis?</i> CPM/scraper library-size norm, log1p, HVG selection, Harmony/scVI batch correction, perturbation-effect normalisation.

B. Pi-Agent Case Study Summary

This short summary highlights four representative cases from the final pi-agent run. Full per-question prompts and stored agent answers are shown in Appendix C.

C. Slash Command Demonstrations

This appendix records the final user-facing outputs for the two PerturbQA slash commands used in evaluation: `/bench [ids...]` and `/paper`. The benchmark results come from `benchmark/results_1780453250096.json`, the best complete pi-agent run.

C.1 `/bench [ids...]` — Final Benchmark Run

```
/bench
```

```
Loaded vector store: 82 documents
```

Table 23: Representative pi-agent behaviours

ID	Topic	Score	Why it matters
gp-001	CRISPR-Cas9 DSB mechanism	89	Single-search factual retrieval: the agent found the exact CRISPR mechanism chunk and covered Cas9 domains, PAM dependence, cleavage position, and repair pathways.
gp-005	MAGeCK pipeline	72	Iterative retrieval recovered a procedural answer that the static pipeline missed: the second query targeted MAGeCK-specific ranking and sgRNA aggregation details.
gp-012	Synthetic lethality	72	Multi-hop analytical retrieval: three searches combined cancer therapy, genetic interaction, and BRCA/PARP context into a grounded answer.
gp-009	CROP-seq vs. Perturb-seq	45	Knowledge-base gap failure: the agent searched effectively, but the corpus lacks a dedicated CROP-seq entry, so the answer remained below the pass threshold.

Running benchmark: 22 questions [pi-agent mode]

Total questions: 22

Passed (≥ 60): 12 (54.5%)

Average score: 62.0/100

Results saved to: benchmark/results_1780453250096.json

The evaluator stores the generated answer preview for each question (`generatedAnswer`, first 500 characters). The table below gives the complete 22-question final run with the question, score, search count, and stored agent answer.

gp-001 — Score 89/100 (PASS)

Question: What is the mechanism by which CRISPR-Cas9 creates double-strand breaks in DNA?

Turns: 2; **Searches:** 1; **Search queries:** CRISPR-Cas9 mechanism double-strand breaks DNA

Agent answer stored by evaluator:

```
## CRISPR-Cas9 Mechanism of Creating Double-Strand Breaks in DNA
```

```
Based on the retrieved knowledge base, here is a comprehensive,
technically precise explanation of the CRISPR-Cas9 double-strand
break (DSB) mechanism:
```

Core Components

The CRISPR-Cas9 system (most commonly using **SpCas9** from *Streptococcus pyogenes*) relies on two essential components:

1. **Cas9 Protein** - A 1,368-amino-acid multidomain endonuclease with the following key functional domains:
 - **RuvC domain**

gp-002 — Score 78/100 (PASS)

Question: How does Perturb-seq combine CRISPR perturbations with single-cell RNA sequencing?

Turns: 2; **Searches:** 2; **Search queries:** Perturb-seq CRISPR single-cell RNA sequencing method | Perturb-seq sgRNA capture transcriptome readout

Agent answer stored by evaluator:

How Perturb-seq Combines CRISPR Perturbations with Single-Cell RNA Sequencing

Perturb-seq is a high-throughput method that couples **pooled CRISPR genetic screens** with **single-cell RNA sequencing (scRNA-seq)**, enabling simultaneous measurement of (1) **which gene was perturbed** (via sgRNA identity) and (2) **the resulting genome-wide transcriptional response** in each individual cell (Dixit et al. 2016). Here is a detailed breakdown of how the integration is achieved:

Core Conc

gp-003 — Score 72/100 (PASS)

Question: What is the difference between CRISPRi and CRISPRa, and when would you use each?

Turns: 3; **Searches:** 3; **Search queries:** CRISPRi CRISPRa gene repression activation comparison | CRISPRi knockdown dCas9 transcriptional interference | CRISPRa activation gene upregulation dCas9 VP64 SAM overexpression screen

Agent answer stored by evaluator:

Great - I've gathered the relevant context from the knowledge base. Here is a comprehensive, technically grounded answer:

CRISPRi vs. CRISPRa: Mechanisms, Differences, and Use Cases

Both CRISPRi (CRISPR interference) and CRISPRa (CRISPR activation) are built on the same foundational component - a **catalytically dead Cas9 (dCas9)** - but they recruit opposite functional effectors to achieve diametrically opposite regulatory outcomes.

The Shared Foundation: Dead Cas9 (dCas9)

gp-004 — Score 62/100 (PASS)

Question: What is GEARS and how does it predict transcriptional responses to genetic perturbations?

Turns: 2; **Searches:** 2; **Search queries:** GEARS predicting transcriptional responses genetic perturbations | GEARS gene interaction graph neural network perturbation

Agent answer stored by evaluator:

GEARS: Gene Expression And Regulation by Similarity

GEARS is a deep learning framework published in *Nature Biotechnology* (2023) that predicts transcriptional responses to **single and combinatorial genetic perturbations** using biological knowledge graphs and graph neural networks (GNNs). It directly addresses the combinatorial explosion problem: exhaustively measuring all gene combination perturbations is intractable (2^n combinations for n genes), so GEARS learns to generalize from

gp-005 — Score 72/100 (PASS)

Question: What are the main steps in the MAGeCK pipeline for analyzing pooled CRISPR screens?

Turns: 2; **Searches:** 2; **Search queries:** MAGeCK pipeline pooled CRISPR screen analysis steps | MAGeCK algorithm sgRNA gene ranking method

Agent answer stored by evaluator:

MAGeCK Pipeline for Analyzing Pooled CRISPR Screens

MAGeCK (Model-based Analysis of Genome-wide CRISPR Knockout) is the standard computational tool for analyzing pooled CRISPR screens. Based on the retrieved knowledge base (*CRISPR Pooled Screens: Design and Analysis*), the pipeline consists of five main steps:

Step 1: Read Count Generation (`mageck count`)

```
The first step maps raw FASTQ sequencing reads to the sgRNA library
and generates a count table.
```

```
```bash
mageck count
```

### gp-006 — Score 78/100 (PASS)

**Question:** What distinguishes a positive selection screen from a negative selection screen in CRISPR pooled screens?

**Turns:** 2; **Searches:** 2; **Search queries:** positive selection negative selection CRISPR pooled screens | CRISPR screen enrichment depletion sgRNA fitness

**Agent answer stored by evaluator:**

```
Positive Selection vs. Negative Selection in CRISPR Pooled
Screens

These two screen types are fundamentally distinguished by their **
biological question**, **the direction of sgRNA abundance change
, and the **selective pressure applied. Here is a detailed
breakdown:

1. Goal / Biological Question

| Screen Type | Goal |
|---|---|
| **Negative Selection (Dropout)** | Identify genes **essential for
cell viability or proliferation** |
| **Positive Selection (Enrichment)** | Iden
```

### gp-007 — Score 52/100 (FAIL)

**Question:** How do genetic interaction scores (epistasis) reveal functional relationships in combinatorial CRISPR screens?

**Turns:** 3; **Searches:** 3; **Search queries:** genetic interaction scores epistasis combinatorial CRISPR screens | combinatorial gene perturbation functional relationships epistasis | synthetic lethality synergistic antagonistic epistasis double knockout screen analysis

**Agent answer stored by evaluator:**

```
Here is a comprehensive, technically grounded answer drawing from
the retrieved literature:

Genetic Interaction Scores (Epistasis) in Combinatorial CRISPR
Screens
```

## ## 1. The Core Concept: Epistasis as Deviation from Additivity

The central logic of epistasis scoring is elegantly simple: **\*\*if two genes are functionally independent, their combined perturbation effect should equal the sum of their individual effects\*\***. Any deviation from this expected additive null reveals a functional rela

### gp-008 — Score 78/100 (PASS)

**Question:** What are off-target effects in CRISPR-Cas9 and how can they be minimized?

**Turns:** 3; **Searches:** 3; **Search queries:** CRISPR-Cas9 off-target effects | minimizing off-target effects CRISPR strategies | high-fidelity Cas9 variants sgRNA specificity guide RNA design off-target cleavage

**Agent answer stored by evaluator:**

Excellent - I now have comprehensive information grounded in the retrieved knowledge base. Here is a detailed, technically rigorous answer:

---

## # Off-Target Effects in CRISPR-Cas9: Causes and Mitigation Strategies

### ## 1. What Are Off-Target Effects?

Off-target effects in CRISPR-Cas9 refer to **\*\*unintended DNA cleavage at genomic sites other than the intended target\*\***. These arise because the standard SpCas9 endonuclease can **\*\*tolerate -15 mismatches\*\*** between the single guide RNA (sgRNA) and t

### gp-009 — Score 45/100 (FAIL)

**Question:** What is CROP-seq and how does it differ from Perturb-seq?

**Turns:** 3; **Searches:** 3; **Search queries:** CROP-seq method CRISPR single-cell RNA sequencing | Perturb-seq pooled CRISPR screens single-cell transcriptomics | CROP-seq sgRNA barcode Pol III transcript Datlinger 2017 single-cell sequencing

**Agent answer stored by evaluator:**

Excellent - the retrieved context provides rich, grounded detail on both methods. Here is a comprehensive technical answer:

---

## ## CROP-seq and Perturb-seq: What They Are and How They Differ

Both **CROP-seq** and **Perturb-seq** are landmark technologies that couple **pooled CRISPR genetic screens** with **single-cell RNA sequencing (scRNA-seq)**. Their shared goal is to simultaneously identify **which gene was perturbed** in each cell and **what transcriptional response** that perturbation caused -

### gp-010 — Score 52/100 (FAIL)

**Question:** How is gene essentiality defined and measured in the context of CRISPR screens?

**Turns:** 3; **Searches:** 3; **Search queries:** gene essentiality definition CRISPR screens | essential genes measurement fitness effects CRISPR knockout | BAGEL2 Bayesian essential gene scoring fitness effect log fold change

**Agent answer stored by evaluator:**

Here is a comprehensive, technically grounded answer based on the retrieved literature:

---

**## Gene Essentiality: Definition and Measurement in CRISPR Screens**

**### 1. Conceptual Definition**

A gene is considered **essential** if its loss causes a significant reduction in cell viability or proliferation. In the context of CRISPR screens, essentiality is an **operationally defined, cell-type-specific, and context-dependent property**: a gene essential in one cell line or condition may be dispensa

### gp-011 — Score 72/100 (PASS)

**Question:** What design principles guide effective sgRNA library construction for genome-wide CRISPR screens?

**Turns:** 3; **Searches:** 3; **Search queries:** sgRNA library design principles genome-wide CRISPR screens | guide RNA library construction pooled CRISPR screen | Brunello GeCKO library sgRNA activity scoring Doench off-target minimization

**Agent answer stored by evaluator:**

The retrieved context is comprehensive. Here is a detailed, technically grounded answer:

---

**# Design Principles for sgRNA Library Construction in Genome-Wide CRISPR Screens**

Effective sgRNA library design is the single most important determinant of screen -qualityit dictates on-target activity,

statistical power, and the interpretability of downstream results. The following principles, drawn from established pooled screen methodology and validated libraries such as **GeCKO v2** and **Brunello**

### gp-012 — Score 72/100 (PASS)

**Question:** What is synthetic lethality and why is it therapeutically relevant in cancer?

**Turns:** 3; **Searches:** 3; **Search queries:** synthetic lethality cancer therapy | synthetic lethal interactions gene perturbation tumor | synthetic lethality BRCA PARP inhibitor DNA repair cancer vulnerability

**Agent answer stored by evaluator:**

Based on the retrieved knowledge, I can now provide a comprehensive , technically grounded answer on synthetic lethality and its therapeutic relevance in cancer.

---

**## Synthetic Lethality: Definition and Therapeutic Relevance in Cancer**

**### What Is Synthetic Lethality?**

**Synthetic lethality** is a genetic phenomenon in which the simultaneous loss-of-function (or inactivation) of **two genes** causes cell death, while the loss of either gene alone is tolerable and does not kill the cell. The t

### gp-013 — Score 52/100 (FAIL)

**Question:** How does base editing differ from prime editing in terms of mechanism and application?

**Turns:** 3; **Searches:** 4; **Search queries:** base editing mechanism application | prime editing mechanism application | base editing cytosine adenine deaminase nickase | prime editing pegRNA reverse transcriptase point mutation insertion deletion

**Agent answer stored by evaluator:**

The PerturbQA knowledge base - which focuses on computational perturbation prediction methods, CRISPR screens, and single-cell genomics - **does not contain dedicated entries on base editing or prime editing** as experimental genome-editing technologies. However, based on the well-established foundational knowledge in the field (including the CRISPR-Cas9 mechanism described in the retrieved context), I can provide a rigorous and comprehensive comparison grounded in the scientific literature.

--

**gp-014 — Score 30/100 (FAIL)**

**Question:** What computational approaches are used to model gene regulatory networks from Perturb-seq data?

**Turns:** 3; **Searches:** 3; **Search queries:** computational approaches gene regulatory networks Perturb-seq | modeling transcriptional regulation single cell CRISPR perturbation data | graph neural network perturbation prediction GEARS gene interaction network

**Agent answer stored by evaluator:**

```
Here is a comprehensive technical overview of the computational
approaches used to model gene regulatory networks (GRNs) from
Perturb-seq data, grounded in the retrieved literature:
```

```

```

```
Computational Approaches for Modeling Gene Regulatory Networks
from Perturb-seq Data
```

```
Perturb-seq data (single-cell RNA-seq coupled with CRISPR
perturbations) offers a uniquely powerful substrate for GRN
inference because each perturbation constitutes a causal
intervention rather than a mere correlation o
```

**gp-015 — Score 72/100 (PASS)**

**Question:** What is the role of the PAM sequence in CRISPR-Cas9 targeting, and what are PAM-less variants?

**Turns:** 2; **Searches:** 2; **Search queries:** PAM sequence CRISPR-Cas9 targeting mechanism | PAM-less CRISPR variants SpRY SpCas9-NG expanded targeting

**Agent answer stored by evaluator:**

```
The retrieved context from the CRISPR-Cas9 Mechanism of DNA
Cleavage paper is comprehensive and directly answers both
parts of your question. Here is a detailed, technically grounded
answer:
```

```

```

```
The PAM Sequence in CRISPR-Cas9 Targeting
```

```
What Is the PAM Sequence?
```

```
The PAM (Protospacer Adjacent Motif) is a short DNA sequence
located immediately downstream (3') of the target DNA site (
protospacer) that is required for Cas9 to recognize and cleave
its target. For the most widely
```

**gp-016 — Score 72/100 (PASS)**

**Question:** How does the CPA (Compositional Perturbation Autoencoder) model predict responses to unseen perturbations?

**Turns:** 2; **Searches:** 2; **Search queries:** CPA Compositional Perturbation Autoencoder unseen perturbations | CPA model architecture latent space drug combination prediction

**Agent answer stored by evaluator:**

```
I now have comprehensive information about CPA from the knowledge
 base. Here is a thorough, technically grounded answer:

How CPA (Compositional Perturbation Autoencoder) Predicts
 Responses to Unseen Perturbations

Reference: Lotfollahi et al., *Molecular Systems Biology*, 2023

Core Idea: Compositional, Disentangled Latent Space

CPA's ability to generalize to unseen perturbations rests on a
 single fundamental principle: **decomposing cellular responses
 into independent, ad
```

**gp-017 — Score 52/100 (FAIL)**

**Question:** What is the Perturb-CITE-seq method and what additional information does it capture over standard Perturb-seq?

**Turns:** 3; **Searches:** 3; **Search queries:** Perturb-CITE-seq method | Perturb-seq standard method single cell CRISPR transcriptomics | Perturb-CITE-seq protein measurement ADT surface protein CRISPR Frangieh 2021

**Agent answer stored by evaluator:**

```
Based on the retrieved knowledge, here is a comprehensive technical
 answer:

Perturb-CITE-seq: Method and Advantages Over Standard Perturb-
 seq

Background: Standard Perturb-seq

Standard **Perturb-seq** (Dixit et al. 2016) couples pooled CRISPR
 genetic screens with **single-cell RNA sequencing (scRNA-seq)**.
 In each cell, it simultaneously captures:
1. **Which gene was perturbed** - via a perturbation barcode (PB)
 linked to the delivered sgRNA, cloned into the 3' UTR of a
 reporter tr
```

**gp-018 — Score 72/100 (PASS)**

**Question:** How is guide RNA efficiency predicted computationally, and what features matter most?

**Turns:** 3; **Searches:** 4; **Search queries:** guide RNA efficiency prediction computational features | CRISPR gRNA on-target activity machine learning model | sgRNA scoring Rule Set 2 Azimuth sequence features nucleotide position | CRISPR guide RNA design off-target prediction tools DeepCRISPR

**Agent answer stored by evaluator:**

```
Computational Prediction of Guide RNA Efficiency: Methods and
Key Features
```

```
The knowledge base contains rich, cross-cutting evidence on this
topic from CRISPR mechanism papers, pooled screen design
literature, and perturbation biology resources. Here is a
comprehensive, technically grounded synthesis.
```

```

```

```
1. What "Guide RNA Efficiency" Means
```

```
Guide RNA (sgRNA) efficiency has two separable dimensions that are
both computationally predicted:
```

```
- **On-target activity**: How effectively does
```

**gp-019 — Score 42/100 (FAIL)**

**Question:** What are the key differences between pooled and arrayed CRISPR screens?

**Turns:** 2; **Searches:** 2; **Search queries:** pooled vs arrayed CRISPR screens differences | CRISPR screen formats pooled arrayed experimental design

**Agent answer stored by evaluator:**

```
Here is a comprehensive comparison of **pooled** vs. **arrayed**
CRISPR screens, grounded in the retrieved knowledge base:
```

```

```

```
Pooled vs. Arrayed CRISPR Screens: Key Differences
```

```
1. Experimental Format
```

```
| Feature | **Pooled Screens** | **Arrayed Screens** |
|---|---|---|
| **sgRNA delivery** | All sgRNAs mixed and delivered together to
one large cell population | Each sgRNA delivered separately, one
per well |
```

```
| **Cell handling** | All perturbations co-exist in the same flask/
culture
```

### gp-020 — Score 52/100 (FAIL)

**Question:** How does scGPT use perturbation data for transfer learning in single-cell foundation models?

**Turns:** 4; **Searches:** 4; **Search queries:** scGPT perturbation transfer learning single-cell foundation model | scGPT fine-tuning gene perturbation prediction | scGPT architecture masked gene modeling perturbation fine-tuning downstream tasks | scGPT bin-based expression tokenization gene perturbation prediction pretraining 33 million cells

**Agent answer stored by evaluator:**

```
The searches have retrieved the direct scGPT capability summary
entry and rich comparative context. Here is a comprehensive,
technically grounded answer:
```

```

```

```
How scGPT Uses Perturbation Data for Transfer Learning
```

```
1. Overview of scGPT as a Foundation Model
```

```
scGPT is a GPT-style (autoregressive/generative transformer) **
single-cell foundation model** pretrained on approximately **33
million human single-cell transcriptomes**. Its design -
philosophy directly analogous to NLP pretraining +
```

### gp-021 — Score 52/100 (FAIL)

**Question:** What is the RNAi mechanism and how does it compare to CRISPR knockdowns for perturbation studies?

**Turns:** 3; **Searches:** 4; **Search queries:** RNAi mechanism gene silencing perturbation | CRISPR knockdown perturbation studies comparison | RNAi siRNA shRNA gene silencing off-target effects knockdown efficiency | CRISPR CRISPRi CRISPRa Perturb-seq single cell perturbation screen

**Agent answer stored by evaluator:**

```
Based on the retrieved context, here is a comprehensive,
technically grounded comparison of RNAi and CRISPR-based
knockdowns for perturbation studies:
```

```

```

```
RNAi Mechanism vs. CRISPR Knockdowns for Perturbation Studies
```

```

```

```
The RNAi Mechanism
```

```
RNA interference (RNAi) is a post-transcriptional gene
silencing mechanism that reduces gene expression at the **mRNA
level** without altering the underlying DNA. The core steps are:
```

```
1. **siRNA/shRNA delivery**: Small interfering RNAs (siRN
```

### gp-022 — Score 45/100 (FAIL)

**Question:** What normalization strategies are commonly applied in single-cell Perturb-seq analysis?

**Turns:** 3; **Searches:** 3; **Search queries:** normalization strategies single-cell Perturb-seq analysis | scRNA-seq normalization preprocessing gene expression perturbation | library size normalization log transformation count data Perturb-seq preprocessing

**Agent answer stored by evaluator:**

```
Excellent - I now have rich, multi-source context to provide a
thorough, grounded answer. Here is a comprehensive technical
overview:
```

```

```

```
Normalization Strategies in Single-Cell Perturb-seq Analysis
```

```
Normalization is a foundational preprocessing step in Perturb-seq
analysis, serving to separate technical variation from true
biological perturbation signals. Based on the literature,
several distinct normalization strategies are commonly applied,
each addressing different sources of noise.
```

```

```

## C.2 /paper — Example: scGPT Foundation Model

```
/paper
selected paper:
2024 scGPT Foundation Model [scgpt-foundation-model]
```

The command opens a paper selector and then evaluates the chosen paper against the six standard PerturbQA capability questions. For `scgpt-foundation-model`, the pre-computed capability matrix is:

Table 24: /paper example output for scGPT Foundation Model

Capability question	Answer	Evidence
Unseen perturbation prediction	Yes	scGPT is fine-tuned for perturbation prediction on Adamson and Norman datasets, achieving competitive performance compared to GEARS and CPA on held-out single-gene perturbations.
Cross cell-line (gene intersection)	Partial	The 33M-cell pre-training corpus spans diverse cell types, enabling transfer via fine-tuning across cell contexts, though explicit cross-cell-line perturbation transfer with gene intersection is not the primary evaluation.
Zero-shot unseen cell line (gene intersection)	Partial	scGPT’s pre-trained representations encode broad cell-state knowledge, enabling partial zero-shot adaptation; however, full perturbation prediction without any fine-tuning on the new cell line is not demonstrated.
Cross technology (gene intersection)	No	scGPT perturbation fine-tuning focuses on CRISPR knockouts; no cross-technology transfer between perturbation modalities is evaluated.
Zero-shot gene misalignment	Yes	scGPT’s masked attention mechanism allows processing arbitrary gene subsets, enabling prediction into different gene vocabularies without re-training on the full gene set.
Perturbation-specificity vs. simple baseline	No	Ahlmann-Eltze et al. (2025) showed that scGPT does not significantly outperform simple additive linear models on Norman and Replogle datasets under well-controlled evaluation metrics.

**Overall capability tier:** Foundation.